Single Level Directory

```c
#include <stdio.h>
#include <string.h>
int main() {
int n, i, j, duplicate;
char files[20][30];
printf("Single Level Directory\n");
printf("Enter number of files: ");
scanf("%d", &n);
for(i = 0; i < n; i++) {
while(1) {
duplicate = 0;
printf("Enter file %d name: ", i + 1);
scanf("%s", files[i]);
for(j = 0; j < i; j++) {
if(strcmp(files[i], files[j]) == 0) {
duplicate = 1;
printf("Error: File name already exists.\n");
printf("Please enter a different file name.\n\n");
break;
}
}
if(duplicate == 0) {
break;
}
}
}
printf("\nFiles in the directory are:\n");
for(i = 0; i < n; i++) {
printf("%s\n", files[i]);
}
return 0;
}
```

```
pratyushojha@Pratyushs-MacBook-Pro Coding % cd "/Users/pratyushojha/Documents/Coding/" && gcc 9a.c -o 9a && "/Users/pratyushojha/Documents/Coding/"9a
1WA24CS216
Single Level Directory
Enter number of files: 3
Enter file 1 name: abcd.txt
Enter file 2 name: pqrs.c
Enter file 3 name: abcd.jpg

Files in the directory are:
abcd.txt
pqrs.c
abcd.jpg
pratyushojha@Pratyushs-MacBook-Pro Coding %
```

Two Level Directory

```
#include <stdio.h>
#include <string.h>
int main() {
int users, files[20];
int i, j, k, duplicate;
char user[20][30];
char filename[20][20][30];
printf("Two Level Directory\n");
printf("Enter number of users: ");
scanf("%d", &users);
for(i = 0; i < users; i++) {
while(1) {
duplicate = 0;
printf("\nEnter username: ");
scanf("%s", user[i]);
for(k = 0; k < i; k++) {
if(strcmp(user[i], user[k]) == 0) {
duplicate = 1;
printf("Error: Username already exists.\n");
printf("Enter a different username.\n\n");
break;}
}
if(duplicate == 0) {
break;
}
}
printf("Enter number of files for %s: ", user[i]);
scanf("%d", &files[i]);
for(j = 0; j < files[i]; j++) {
while(1) {
duplicate = 0;
printf("Enter file %d name: ", j + 1);
scanf("%s", filename[i][j]);
for(k = 0; k < j; k++) {
if(strcmp(filename[i][j], filename[i][k]) == 0) {
duplicate = 1;
printf("Error: File already exists for this user.\n");
printf("Enter a different filename.\n\n");
break;
}
}
if(duplicate == 0) {
break;
}
}
}
}
printf("\n\nDirectory Structure:\n");
for(i = 0; i < users; i++) {
```

```
printf("\nUser: %s\n", user[i]);
for(j = 0; j < files[i]; j++) {
printf("%s\n", filename[i][j]);
}
}
return 0;
}
```

```
pratyushojha@Pratyushs-MacBook-Pro Coding % cd "/Users/pratyushojha/Documents/Coding/" && gcc 9b.c -o 9b && "/U
sers/pratyushojha/Documents/Coding/"9b
1WA24CS216
Two Level Directory
Enter number of users: 2

Enter username: Documents
Enter number of files for Documents: 2
Enter file 1 name: project.txt
Enter file 2 name: notes.txt

Enter username: College
Enter number of files for College: 2
Enter file 1 name: project.txt
Enter file 2 name: notes.txt


Directory Structure:

User: Documents
project.txt
notes.txt

User: College
project.txt
notes.txt
pratyushojha@Pratyushs-MacBook-Pro Coding %
```

Hierarchical Directory

```
#include <stdio.h>
#include <string.h>
int main() {
int mainDir, subDir[20], files[20][20];
int i, j, k, l;
int duplicate;
char mainDirectory[20][30];
char subDirectory[20][20][30];char filename[20][20][20][30];
printf("Hierarchical Directory\n");
printf("Enter number of main directories: ");
scanf("%d", &mainDir);
for(i = 0; i < mainDir; i++) {
while(1) {
duplicate = 0;
printf("\nEnter main directory name: ");
scanf("%s", mainDirectory[i]);
for(l = 0; l < i; l++) {
if(strcmp(mainDirectory[i], mainDirectory[l]) == 0) {
duplicate = 1;
printf("Error: Main directory already exists.\n");
printf("Enter a different directory name.\n\n");
break;
}
}
if(duplicate == 0) {
break;
}
}
printf("Enter number of subdirectories in %s: ", mainDirectory[i]);
scanf("%d", &subDir[i]);
for(j = 0; j < subDir[i]; j++) {
while(1) {
duplicate = 0;
printf("Enter subdirectory name: ");
scanf("%s", subDirectory[i][j]);
for(l = 0; l < j; l++) {
if(strcmp(subDirectory[i][j], subDirectory[i][l]) == 0) {
duplicate = 1;
printf("Error: Subdirectory already exists.\n");
printf("Enter a different subdirectory name.\n\n");
break;
}
}
if(duplicate == 0) {
break;
}
}printf("Enter number of files in %s: ", subDirectory[i][j]);
scanf("%d", &files[i][j]);
for(k = 0; k < files[i][j]; k++) {
while(1) {
duplicate = 0;
printf("Enter file %d name: ", k + 1);
```

```
scanf("%s", filename[i][j][k]);
for(l = 0; l < k; l++) {
if(strcmp(filename[i][j][k], filename[i][j][l]) == 0) {
duplicate = 1;
printf("Error: File already exists.\n");
printf("Enter a different filename.\n\n");
break;
}
}
if(duplicate == 0) {
break;
}
}
}
}
}
printf("\n\nHierarchical Directory Structure:\n");
for(i = 0; i < mainDir; i++) {
printf("\n%s\n", mainDirectory[i]);
for(j = 0; j < subDir[i]; j++) {
printf(" %s\n", subDirectory[i][j]);
for(k = 0; k < files[i][j]; k++) {
printf(" %s\n", filename[i][j][k]);
}
}
}
return 0;
}
```

```
pratyushojha@Pratyushs-MacBook-Pro Coding % cd "/Users/pratyushojha/Documents/Coding/" && gcc 9c.c -o 9c && "/U
sers/pratyushojha/Documents/Coding/"9c
1WA24CS216
Hierarchical Directory
Enter number of main directories: 1

Enter main directory name: Project
Enter number of subdirectories in Project: 2
Enter subdirectory name: abc
Enter number of files in abc: 1
Enter file 1 name: abcd
Enter subdirectory name: 1
Enter number of files in 1: 2
Enter file 1 name: pra
Enter file 2 name: prat


Hierarchical Directory Structure:

Project
  abc
    abcd
  1
    pra
    prat
pratyushojha@Pratyushs-MacBook-Pro Coding %
```